

ODIN Improvement Lab Notebook (Initial Session)

Dataset and Setup

- **T-1 dataset:** `ODIN_ENRICHED_PDUFA_1933_v4_T1_COMPLIANT.csv` with 1 933 events and 33 columns. Each row contains metadata (ticker, company, therapeutic area, PDUFA date) and labels (`APPROVAL` vs `CRL`).
- **Engine version:** the provided `odin_v104.py` implements **v10.5** with additive weights, optional isotonic calibrator, and tiering. The code uses a lower base rate (60.8 %) and different signal weights compared to the `SKILL.md` spec for v10.4.
- **Files:** previous optimization outputs `odin_sniper_v15_best_config.json` and `odin_sniper_v15_final.csv` were inspected for future reference. They show an alternative parameter search with additional extended features (sponsor-profile, modality, interaction terms, year buckets etc.).

Baseline Performance of v10.5 on T-1 Dataset

All signals requiring external MCP data (S6 Hiring, S12 EMA CMC flags, S22 Pediatric PK, S23 Insider, social signals) were absent in the T-1 dataset; hence the evaluation reflects core additive signals (designations, AdCom, CRL, sponsor experience, therapeutic area) and optional penalties. No external calibrator (`odin_v105_calibrator.pkl`) was available, so raw scores were used.

- **Overall metrics** (1 933 events):
 - Accuracy at $p \geq 0.5$ threshold: **87.1 %**
 - ROC AUC: **0.838**
 - Brier score (uncalibrated): **0.152**
 - Log loss: **0.485**
- Confusion matrix (Threshold 0.5):
 - True positives: 1 600
 - False positives: 249
 - False negatives: 0
 - True negatives: 84
- **Tier-wise hit rates:** the tier cutoffs were 0.655/0.739/0.840 (v10.5 defaults). The table below shows the number of events and the proportion of approvals within each tier.

Tier	Count	Approval rate
TIER 1	48	1.00
TIER 2	136	0.97

Tier	Count	Approval rate
TIER 3	594	0.96
TIER 4	1 155	0.74

The engine places **83 %** of events into TIER 3–4 and uses only a few events to differentiate between Tiers 1–2. Tier thresholds may therefore be too lenient; most true approvals fall in lower tiers.

- **Calibration:** decile-wise reliability shows systematic underestimation for high-probability bins. For example, the highest decile (mean predicted ≈ 0.82) has an observed approval rate of ≈ 0.98 , while mid-range deciles (predicted ≈ 0.60 – 0.70) have actual approval rates 0.87–0.97.
- **Bias diagnostics:** analysis by therapeutic area and sponsor experience highlighted consistent under-prediction for some categories:
- **Infectious Disease & Oncology:** true approval rates ≈ 89 %; mean predicted probabilities only 0.69–0.72.
- **Experienced sponsors** (≥ 5 prior approvals) had 95 % approval, but v10.5 predicted ≈ 0.67 . Inexperienced sponsors (0 prior approvals) had ~ 67 % approval with predicted ≈ 0.59 .
- Thus, the base rate and some weight scalars may be too conservative.

Logistic Calibration Experiment

To improve calibration without altering the additive structure, a simple logistic regression was trained to map raw scores to calibrated probabilities. A 5-fold cross-validation on the T-1 dataset produced:

- Mean ROC AUC: **0.838** (unchanged from the raw model).
- Mean Brier score: **0.103** vs **0.152** for the raw model.
- Mean accuracy (threshold 0.5): **87.0 %** (similar).

The logistic model effectively steepened the probability curve (intercept ≈ -4.74 , coefficient ≈ 10.3) and reduced miscalibration. A simple isotonic or logistic calibrator could therefore be added as a second-layer to the existing engine. This calibration layer is easy to maintain interpretability since it acts on the aggregate score.

Identified Limitations and Improvement Opportunities

1. **Underestimation of high-quality events:** experienced sponsors and high-impact therapeutic areas have higher observed approval rates than predicted. Adjusting the base rate upward (e.g., from 60.8 % back towards the v10.4 base of 82.7 %) or increasing the TA adjustment weight would better reflect recent approval trends.
2. **Over-concentration in lower tiers:** the current tier thresholds (0.655/0.739/0.840) push 60 % of events into TIER 4 and only 10 % into TIER 1–2. For actionable trading guidance, more discriminatory thresholds (e.g., using percentile cuts 70/80/90 %) or dynamic thresholds per therapeutic area may improve hit-rate segmentation.

3. **Lack of interaction terms:** the additive model cannot capture interactions such as TA × prior CRL or sponsor × designation synergy. Piecewise or bucketed weights (e.g., different CRL penalty for oncology vs pain, or enhanced Orphan + Priority boost) could be explored while keeping interpretability.
4. **Missing signals:** the dataset lacks MCP-derived inputs (S6 hiring, S12 EMA CMC, S22 pediatric PK, S23 insider selling, S17–S21 social). Real-world deployment must ensure these signals are populated; otherwise, a completeness/confidence score should flag lower confidence.
5. **Calibration layer:** adding a logistic or isotonic calibration layer on the aggregated raw score (Layer 2) meaningfully improves probability accuracy without sacrificing interpretability. This calibrator should be trained on a rolling out-of-sample window to avoid overfitting.

Next Steps and Research Backlog

1. **Parameter tuning search** (Track C):
2. Refactor the engine into a vector of tunable parameters and implement a vectorized evaluator (batch-wise evaluation on GPU). Constrain weights/penalties to plausible ranges and enforce monotonic tier thresholds.
3. Explore discrete grids around v10.5 defaults; record metrics such as Brier score, AUC, Tier-1 coverage/hit rate. Keep a leaderboard of top configurations.
4. **Refined TA and sponsor adjustments** (Track B):
5. Use the T-1 dataset to fit per-TA correction factors. For example, Infectious Disease and Oncology could have positive offsets added to the TA lookup table, while Pain/Ophthalmology may warrant stronger penalties.
6. Adjust sponsor experience thresholds and weights to better align with observed 95 % vs 67 % approval rates.
7. Investigate separate CRL penalties for different modalities or therapeutic areas.
8. **Confidence/Completeness scoring** (Track F):
9. Define a confidence metric based on available signal blocks (MCP signals vs missing). For example, assign HIGH confidence when all core signals (S1–S23) are present, MEDIUM when only designations/AdCom/CRL/sponsor/TA are available, and LOW when multiple blocks missing. Use this to adjust trading guidance or provide cautionary notes.
10. **Calibration layer implementation** (Track D):
11. Integrate a logistic calibrator into the engine (Layer 2), and persist the fitted model (e.g., as `odin_v105_calibrator.pkl`). Test isotonic regression as well.
12. Evaluate performance on a time-based hold-out (e.g., train on pre-2024 data, test on 2024–2025 events) to simulate forward deployment.
13. **Investigate new signals** (Track E):

14. Consider sponsor-level financial metrics (leverage, cash runway) and historical CRL types.
 15. Incorporate clinical trial design features (endpoint type, surrogate vs hard outcome, enrollment size) by leveraging the MCP ClinicalTrials connector.
 16. Explore enriched CMC/manufacturing flags using CMO risk profiles (e.g., CDMO tiering, modality complexity). The `modality` column in the dataset hints at this direction.
 17. Quantify the incremental AUC/Brier improvement of each candidate signal using backtests.
 18. **Robustness testing** (Track F):
 19. Simulate edge scenarios such as pediatric sNDAs with missing PK data, conflicting signals (e.g., strong hiring vs EMA CMC flag), or unknown sponsors/therapeutic areas. Ensure the engine gracefully defaults to base scores and flags lower confidence.
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